

# Distributed Control and Communication Framework for Connectivity Preservation in Flow-Driven Robot Fleets

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**Abstract**—We study the deployment of underwater robot fleets in large, flow dominated environments where localization is denied, communication is limited, and energy is scarce. The robots need to travel long distances, detect emerging targets, and stay connected using only local information. To meet these challenges we design a distributed framework that exploits natural drift and activates control only when required. The method begins with the robots deployed as a fully connected network. Using minimal information exchange, each robot prunes non-critical edges, producing a sparse and tree like structure that spans the entire team. Upon this structure we construct a distributed control Lyapunov function–control barrier function (CLF–CBF) framework. Each robot enforces barrier constraints to maintain the necessary connections, while redundant links are discarded when alternative paths emerge. When a target is discovered, this framework ensures the detecting robot attends the target, while the team reconfigures to provide a connected formation. This approach guarantees long term connectivity with very low control effort, and enables collective response to dynamic tasks. Simulations validate the framework and demonstrates that it yields spanning structures that converge toward a minimum spanning tree over time, highlighting both its efficiency and robustness.

## I. INTRODUCTION

We are interested in deploying a fleet of underwater robots into large-scale, challenging environments such as the deep ocean and under-ice cavities, where localization is denied, communication bandwidth is limited, and the environment is dominated by dynamic flows and eddies [1]. These robots must traverse long distances on limited energy budgets, discover emerging regions of interest, sample the environment, and relay data, all while relying on intermittent and bandwidth-limited communication [2], [3].

The central challenge is to design strategies that enable the fleet to (i) efficiently cover a large spatial domain to discover emerging targets while maintaining network connectivity, and (ii) realize a globally connected configuration using only localized information and limited communication.

Our proposed approach is to deploy the robots together and then allow them to drift with the prevailing flow. Control effort is only applied when it is necessary. This includes situations when the robots risk drifting outside a prescribed region or when they risk losing connectivity. In this way the formation disperses while staying connected, and the group

can also reconfigure when a new target location is discovered or generated [4]. This strategy is bold, but it is supported by recent progress. Previous work has shown that surface vehicles can drift with the flow while using minimal control to remain in gyre like structures [5], or track the periodicity of revisits via intermittent control triggered by prescribed conditions [6]. In addition, broadly adopted models describe the uncontrolled motion of objects in water as drift combined with Brownian type diffusion [7]–[9], which align with observation on underwater robots [10], [11]. This indicates that our fleet will approximately follow the main direction of the flow while also dispersing stochastically.

Although dispersion promotes coverage of large task spaces, two challenges remain open. The first is how to maintain connectivity. The second is how to enable re-configuration around targets while working with unreliable communication that cannot support centralized control. Prior work on connectivity maintenance and target surrounding has emphasized the importance of graph theoretic methods [12], [13]. Control barrier functions (CBFs) have received attention for their energy efficiency in this setting. For example, CBFs have been applied to the Fiedler value of the network graph in order to maintain connectivity [14]. This approach is effective but it requires centralized and computationally expensive eigenvalue calculations. Other work has applied distributed CBF formulations that keep robots pairwise within communication range [15], [16]. More recently, a hybrid framework was introduced that combines distributed CBFs with a graph theoretic tool, where each robot interacts only with a limited set of neighbors defined by a minimum spanning tree (MST) [17]. This solution is promising since each robot only applies CBF to itself and to selected neighbors, but it still requires generating a minimum spanning tree that depends on global information.

In our case a feasible framework must combine two ingredients. The first is a distributed neighbor selection strategy so that each robot chooses a limited set of neighbors to remain within communication range. The second is a local control law applied only to these neighbors. Together these two features must guarantee global connectivity. Work on spanning trees has shown that distributed verification methods exist, but the actual construction of a minimum spanning tree is classically centralized [18]–[20]. Progress in fully distributed protocol for exact MST construction [21], [22] remain communication-intensive. As a result, several approximate or near-optimal distributed methods have been proposed, but they do not achieve exact solutions [23].

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We take advantage of two physical features of our problem. First, the robots are deployed together, and the initial configuration forms a fully connected graph. Second, the purpose of a spanning tree is to reduce long range connections. A robot should drop links to distant neighbors whenever a closer neighbor can maintain connectivity at lower control cost. Recent progress [24] has also shown that robots can verify whether an edge is critical, meaning that breaking it would break the network connectivity, using distributed procedures that rely only on very limited data sharing. Based on these observations we propose a distributed method that requires very little information exchange. Each robot shares only a list of its neighbor indices. With this limited data the robots identify and prune edges that are not critical to connectivity. But unlike demonstrated in [24], we are introducing an algorithm which can identify and prune the edges concurrently rather than a traditional sequential approach still ensuring maintaining connectivity between all the robots. The result is a sparse and tree-like structure that spans the entire network. Once this structure is established, we construct a distributed control Lyapunov function–control barrier function (CLF–CBF) framework upon it. Each robot maintains only the necessary connections by enforcing CBF constraints, and redundant links are pruned whenever new edges provide alternative paths. When a target is detected, the framework drives the robot that discovers it toward the target, while the rest of the team reconfigures to form a supporting network from the bottom up. As a result, the team preserves connectivity, expands to maintain coverage, and collectively shifts toward the target location.

Simulations demonstrate that this method can efficiently identify the critical edges and prune the non critical ones while reaching a target location of the environment and keep the network connected. Over deployment time the spanning structure produced by this distributed pruning strategy converges toward a minimum spanning tree. This shows both its long term efficiency and its robustness in flow dominated and resource constrained conditions.

This paper is organized as follows. Sec. II introduces the problem. Sec. III presents our methodology of achieving the structure and moving towards the desired location. Sec. IV certifies the preservation of the communication links over the graph. Sec. V explains the experimental setup and Sec. VI demonstrates the results of the proposed method via simulated results. Sec. VII concluded this work.

## II. BACKGROUND AND PROBLEM FORMULATION

We now formalize the problem. This section introduces the robot dynamics model, states the assumptions, defines the graph-theoretic notation, and presents the formal problem statement.

### A. Advection-diffusion model [7]–[9]

We model a fleet of  $N$  robots operating in  $\mathbb{R}^n$  (with  $n = 2$ , used for visualization in our simulations) released together in the water, their collective motion can be decomposed into a drift component induced by an ambient flow field and a

stochastic dispersion component due to unexpected turbulence and small perturbations. Let  $x_i(t) \in \mathbb{R}^n$  denote the position of the  $i$ -th robot at time  $t$ , where  $i \in \{1, 2, \dots, N\}$ , and let

$$\bar{x}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t)$$

be the center of mass of the group.

The evolution of the center of mass is governed primarily by the advective flow field, i.e.,

$$\dot{\bar{x}}(t) \approx f_{\text{flow}}(\bar{x}(t), t),$$

where  $f_{\text{flow}}$  is the background velocity field. Each individual object then deviates from  $\bar{x}(t)$  due to unresolved small-scale turbulence and fluctuations, which can be modeled as Brownian motion:

$$\begin{aligned} dx_i(t) &= (f_{\text{flow}}(x_i(t), t) + v_i(t))dt + \sigma_{\text{diff}}dW_i(t) \\ \dot{v}_i(t) &= u_i(t) - \gamma v_i(t) \end{aligned} \quad (1)$$

where  $v_i(t) \in \mathbb{R}^n$  is the velocity of each individual robot,  $u_i(t) \in \mathbb{R}^n$  is the thrust,  $\gamma > 0$  is a velocity damping factor, and  $W_i(t)$  is a Wiener process capturing diffusion (with intensity  $\sigma_{\text{diff}}$ ).

Thus the group drifts together with the flow through the motion of  $\bar{x}(t)$ , while individuals spread stochastically around the drifting center due to the independent diffusion terms.

As an example, the flow field is a base current plus a spatially varying swirl:

$$f_{\text{flow}}(x, t) = v_{\text{base}}(t) + A_{\text{swirl}} \begin{bmatrix} \sin\left(\frac{x_2 + t}{s_{\text{swirl}}}\right) \\ \cos\left(\frac{x_1 + 0.3t}{s_{\text{swirl}}}\right) \end{bmatrix}$$

For two robots  $i$  and  $j$ , the flow difference is generally nonzero when  $x_i \neq x_j$ :

$$f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t) = A_{\text{swirl}} \begin{bmatrix} \sin\left(\frac{x_{i,2} + t}{s_{\text{swirl}}}\right) - \sin\left(\frac{x_{j,2} + t}{s_{\text{swirl}}}\right) \\ \cos\left(\frac{x_{i,1} + 0.3t}{s_{\text{swirl}}}\right) - \cos\left(\frac{x_{j,1} + 0.3t}{s_{\text{swirl}}}\right) \end{bmatrix} \neq 0$$

We assume that the ambient flow is spatially smooth and its differential drift is bounded:

$$\|f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)\| \leq L_f \|x_i - x_j\|, \quad \forall x_i, x_j, \quad \forall t \geq 0.$$

If the flow were spatially uniform, it would cancel in the relative dynamics. This assumption explicitly captures the non-uniform, spatially varying regime by bounding how much the flow can differ across inter-robot distances. For the swirl flow field, one can take this flow as globally lipschitz in  $x$  with a lipschitz constant  $L_f \approx A_{\text{swirl}}/s_{\text{swirl}}$ .

### B. Control-affine model

For controller synthesis and theoretical analysis, we now abstract the above simulation dynamics into a compact control-affine model on  $x_i$ :

$$\dot{x}_i(t) = f_{\text{flow}}(x_i(t), t) + u_i(t) + d_i(t), \quad \|d_i(t)\| \leq \bar{d}. \quad (2)$$

With the control input  $u_i(t)$  set to zero throughout the deployment time window, the robots will be passively advected by the background flow while gradually dispersing due to stochastic perturbations. Here the unmodeled effects of drag, turbulence and stochastic perturbations are captured by a bounded disturbance term:

$$\|d_i(t)\| \leq \bar{d}, \quad \forall i \in \mathcal{V}, \quad \forall t \geq 0.$$

This term corresponds to the combined effect of Brownian motion, transient velocity dynamics, and model uncertainties that are abstracted away in the reduced model.

We present two different dynamics descriptions because they serve different purposes. The advection-diffusion model supports physical interpretation and simulation realism, while the control affine model is used to controller design and theoretical analysis in section IV. The flow term is evaluated at each robot's position, so it is not a common input that is identical for all the robots in relative coordinates  $z_{ij} = x_i - x_j$ ,

$$\dot{z}_{ij} = (f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)) + (u_i - u_j) + (d_i - d_j)$$

So, the flow contribution cancels only in the special case of spatially uniform flow. In our setting, spatial variation of  $f_{\text{flow}}$  is explicitly bounded, which is a quantity that enters the connectivity and invariance analysis.

The time derivative of the squared inter-robot distance can be written as

$$\dot{z}_{ij}^2 = 2z_{ij}^\top \left( (f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)) + (v_i - v_j) \right)$$

This expression makes explicit that the flow contributes through the *difference term*  $f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)$ .

Notice that the choice of a two-dimensional environment is made solely for ease of visualization. The developed method is dimension-agnostic.

### C. Assumptions on communication, control and connectivity

**Assumption 1** (Bounded actuation and bounded exogenous effects). *At time  $t$ , the communication network is an undirected disk graph  $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$  with range  $R_{\max} > 0$ :*

$$(i, j) \in \mathcal{E}_t \iff \|x_i(t) - x_j(t)\| \leq R_{\max}. \quad (3)$$

*Each robot  $i$  communicates only with its neighbors in its neighboring set*

$$\mathcal{N}_i(t) \triangleq \{j \in \mathcal{V} : \|x_i(t) - x_j(t)\| \leq R_{\max}\}, \quad (4)$$

*using only neighbor communication without any central coordination. Each robot has a unique identifier to index neighbor messages consistently. The initial graph  $\mathcal{G}_0$  is connected.*

*For each robot  $i$  and all  $t \geq 0$ , the control is bounded as  $\|u_i(t)\| \leq u_{\max}$ . The unmodeled disturbance is bounded by  $\|d_i(t)\| \leq \bar{d}$ . The ambient flow field  $f_{\text{flow}}(x, t)$  is Lipschitz in space with constant  $L_f$  over the operating domain, i.e.,  $\|f_{\text{flow}}(x, t) - f_{\text{flow}}(y, t)\| \leq L_f \|x - y\|$  for all relevant  $x, y, t$ .*

**Assumption 2** (Feasible safety & connectivity margins). *There exist margins  $\varepsilon_s > 0$  and  $\varepsilon_c > 0$  such that the required*

*inter-robot separation and communication constraints are simultaneously feasible:*

$$0 < d_{\min} + \varepsilon_s < R_{\max} - \varepsilon_c.$$

*Moreover, the available control authority is sufficient to counter worst-case relative disturbance over the communication range, i.e.,*

$$u_{\max} > \bar{d} + L_f R_{\max}.$$

**Assumption 3** (Time-scale separation for distributed updates). *The distributed pruning updates run on a faster time scale than the robot motion, so that during one algorithmic iteration the neighbor set can be treated as constant and required neighbor-to-neighbor messages are available for that iteration. Equivalently, topology variations are slow relative to the update period.*

### D. Problem Statement

Consider the multi-robot system introduced above with dynamics (2) and communication model (3)–(4). The robots are advected by a spatially varying ambient flow field  $f_{\text{flow}}(x, t)$  and are subject to bounded disturbances  $d_i(t)$ , which together induce natural dispersion. While this dispersion promotes exploration, it can also stretch inter-robot distances and break communication links.

Let  $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$  denote the physical disk graph induced by the range  $R_{\max}$  (Assumption 1). During the mission, robots may intentionally sparsify the network by maintaining a time-varying coordination subgraph  $\mathcal{G}_{\text{topo}}(t) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(t))$  with  $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$ .

We formalize the hard safety/connectivity requirements using the constraint sets

$$\begin{aligned} \mathcal{C}_{\text{safe}} &\triangleq \left\{ x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \geq d_{\min}, \forall i \neq j \right\}, \\ \mathcal{C}_{\text{conn}}(\mathcal{E}) &\triangleq \left\{ x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \leq R_{\max}, \forall (i, j) \in \mathcal{E} \right\}, \end{aligned}$$

and the admissible input set  $\mathcal{U} \triangleq \{(u_1, \dots, u_N) : \|u_i\| \leq u_{\max} \forall i\}$  (Assumption 1).

Targets are not known a priori and are detected only when an agent enters its sensing range. During exploration, the team should drift with  $f_{\text{flow}}$  while remaining connected and reducing communication redundancy. Upon target detection, agents should initiate goal-seeking behavior maintaining end-to-end connectivity across the fleet.

Each agent  $i$  has access only to local information: (i) its own state  $x_i$ ; (ii) relative states  $\{x_j - x_i\}_{j \in \mathcal{N}_i(t)}$  and messages from one-hop neighbors; and (iii) local graph quantities, like two hop connection, are computable through neighbor exchanges. We denote this information set by  $\mathcal{I}_i(t)$ . No centralized computation, global maps, spanning trees, or spectral quantities are assumed.

Distributed connectivity-preserving control with topology sparsification. Design distributed policies consisting of (i) a feedback control law  $u_i(t) = \kappa_i(\mathcal{I}_i(t))$  for each agent and (ii) a distributed concurrent pruning rule that generates  $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$ , using only local information, such that for all  $t \geq 0$ :

- 1)  $x(t) \in \mathcal{C}_{\text{safe}}$  and  $x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_{\text{topo}}(t))$ , and the maintained graph  $\mathcal{G}_{\text{topo}}(t)$  remains connected.
- 2) The controller exploits advection and uses minimal actuation.
- 3) The team spreads for coverage during exploration and, while expansion figuring out the current graph structure from local knowledge and concurrently detecting and pruning non-critical edges to make a sparse graph network.

The proposed framework in Sec. III provides a fully distributed realization of these objectives via distributed topology sparsification and decentralized control.

### III. METHODOLOGY

Our framework couples a *graph layer* with a *control layer*. The graph layer executes a fully distributed *adjacency-consensus and concurrent pruning* routine that reduces redundant links while preserving end-to-end connectivity. The control layer implements a CLF–CBF safety filter: a nominal CLF controller drives robots toward task goals, while CBF constraints enforce (i) collision avoidance with nearby robots and (ii) connectivity range maintenance for critical communication edges.

**Notation.** Robots evolve under the control-affine model (2). The instantaneous communication disk graph is defined in (3)–(4). At pruning iteration  $k$ , let  $\mathcal{E}(k)$  denote the set of *active* communication links. For robot  $i$ , define the *active neighbor set*  $\mathcal{T}_i(k) \triangleq \{j : (i, j) \in \mathcal{E}(k)\}$ . Separately, define a *safety neighbor set*  $\mathcal{S}_i(t)$  used only for collision avoidance.

#### A. Graph layer: adjacency consensus with concurrent pruning

Each robot  $\ell$  maintains a local estimate  $A^\ell(k) \in \mathbb{R}^{N \times N}$  of the adjacency matrix [24]. At every iteration  $k$ , robot  $\ell$  performs a consensus update using only 1-hop exchanges with  $\mathcal{N}_\ell(k)$ , followed immediately by a three-phase pruning step.

1) *Adjacency matrix consensus update:* Robot  $\ell$  updates

$$A^\ell(k+1) = A^\ell(k) + T_d \Delta A^\ell(k), \quad (5)$$

$$\Delta A_{ij}^\ell(k) = \sum_{p \in \mathcal{N}_\ell(k)} w_{lp} \left( A_{ij}^p(k) - A_{ij}^\ell(k) \right) + \varepsilon_{ij}^\ell(k), \quad (6)$$

where  $T_d > 0$  is a stepsize,  $w_{lp}$  are averaging weights and  $\varepsilon_{ij}^\ell(k)$  injects local observations for edges incident to  $\ell$ . A common choice is an observation “trust” signal

$$t_{ij}^\ell(k) = \exp\left(-\frac{d_{ij}^2(k)}{\sigma^2}\right), \quad (7)$$

built from measured distance  $d_{ij}(k)$  and parameter  $\sigma > 0$ , and then setting  $\varepsilon_{ij}^\ell(k)$  so that  $A_{ij}^\ell$  is nudged toward  $t_{ij}^\ell(k)$  for incident edges and unchanged otherwise.

2) *Local progress metric and adaptive thresholds:* To prevent aggressive pruning before estimates stabilize, each robot tracks a local disagreement energy, e.g.,

$$V_\ell(k) \triangleq \sum_{p \in \mathcal{N}_\ell(k)} \|A^\ell(k) - A^p(k)\|_F^2, \quad (8)$$

and uses *adaptive* schedules that transition from conservative to aggressive pruning:

$$\varepsilon(k) = \varepsilon_{\max} V_\ell(0) \exp\left(-\alpha \frac{k}{k_{\max}}\right) + \varepsilon_{\min} V_\ell(0), \quad (9)$$

$$\mu(k) = \mu_{\max} \exp\left(-\beta \frac{k}{k_{\max}}\right) + \mu_{\min}. \quad (10)$$

Here  $\varepsilon(k)$  enforces minimum consensus progress per pruning step, and  $\mu(k)$  is a safety margin for algebraic connectivity checks.

3) *Three-phase concurrent pruning protocol:* At the same iteration  $k$ , robots prune edges while consensus continues:

- **Phase 1 (Proposal):** robot  $\ell$  locally evaluates candidate incident edges  $e = (\ell, j)$  for removal using only  $A^\ell(k)$  and neighbor messages. A candidate is proposed only if it satisfies: (i) an *alternative path* exists between  $\ell$  and  $j$  after removing  $e$ , (ii) the resulting graph is predicted connected, (iii) the predicted algebraic connectivity satisfies  $\lambda_2 > \lambda_{\min} + \mu(k)$ , and (iv) a *progress guard* holds, e.g., a one-step simulated update predicts  $V_\ell(k+1)_{\text{sim}} < V_\ell(k) - \varepsilon(k)$ .
- **Phase 2 (Negotiation):** edge  $(i, j)$  may be removed only with *bilateral consent*; the endpoints exchange accept or reject messages.
- **Phase 3 (Conflict resolution):** to avoid inconsistent simultaneous removals, robots apply a deterministic selection rule so that only a non-conflicting set of edges is removed at iteration  $k$ . Practical tie-breakers prioritize pruning longer edges.

The output of the graph layer is the updated *active* edge set  $\mathcal{E}(k+1)$  and corresponding active neighbor sets  $\mathcal{T}_i(k+1)$ .

#### B. Control layer: CLF goal-seeking with CBF safety & critical-edge maintenance

Between pruning iterations, each robot runs a CLF–CBF controller under (2). The CLF term drives robots toward task goals, while CBF constraints enforce: (i) *collision avoidance* for all  $j \in \mathcal{S}_i(t)$ , and (ii) *range maintenance* for all  $j \in \mathcal{T}_i(k)$ .

1) *Barrier functions (pairwise safety and connectivity):* For any ordered pair  $(i, j)$  define  $z_{ij} \triangleq x_i - x_j$  and  $r_{ij} \triangleq \|z_{ij}\|$ . We use the standard squared-distance barriers

$$h_{ij}^{\text{safe}}(x) \triangleq r_{ij}^2 - d_{\min}^2, \quad (11)$$

$$h_{ij}^{\text{conn}}(x) \triangleq R_{\max}^2 - r_{ij}^2, \quad (12)$$

so that  $h_{ij}^{\text{safe}} \geq 0$  implies  $\|x_i - x_j\| \geq d_{\min}$  and  $h_{ij}^{\text{conn}} \geq 0$  implies  $\|x_i - x_j\| \leq R_{\max}$ .

To handle spatially varying flow and bounded disturbances (Assumptions in Sec. II), we employ linear CBF inequalities.

With extended class- $\mathcal{K}$  functions  $\alpha_s, \alpha_c$ , we enforce for each  $(i, j) \in \mathcal{S}$ :

$$2z_{ij}^\top(u_i - u_j) \geq 2L_f r_{ij}^2 + 4\bar{d}r_{ij} - \alpha_s(h_{ij}^{\text{safe}}(x)), \quad (13)$$

and for each  $(i, j) \in \mathcal{E}(k)$ :

$$2z_{ij}^\top(u_i - u_j) \leq \alpha_c(h_{ij}^{\text{conn}}(x)) - 2L_f r_{ij}^2 - 4\bar{d}r_{ij}. \quad (14)$$

In implementation, each robot uses the most recently received neighbor control  $u_j$  to render these constraints affine in its own decision variable  $u_i$ .

2) *CLF objective and QP-based safety filter*: Let  $x_i^{\text{goal}}(t)$  denote robot  $i$ 's goal reference. We adopt the quadratic CLF

$$V_i(x_i) \triangleq \|x_i - x_i^{\text{goal}}(t)\|^2, \quad (15)$$

and a nominal goal-seeking command  $u_i^{\text{nom}}(t)$ . Robot  $i$  then computes a minimally modifying control via a quadratic program:

$$u_i^*(t) = \arg \min_{u_i, \delta_i} \|u_i - u_i^{\text{nom}}\|^2 + \rho \delta_i^2 \quad (16)$$

$$\text{s.t. } \dot{V}_i(x_i) \leq -cV_i(x_i) + \delta_i \quad (17)$$

$$(13) \quad \forall j \in \mathcal{S}_i(t), \quad (14) \quad \forall j \in \mathcal{T}_i(k), \quad (18)$$

$$\|u_i\| \leq u_{\max}. \quad (19)$$

The relaxation  $\delta_i$  preserves feasibility when hard constraints dominate;  $\rho$  penalizes relaxation. Crucially, connectivity CBFs are enforced only for the *active* neighbor set  $\mathcal{T}_i(k)$ , so the number of connectivity constraints scales with the pruned topology rather than the dense disk graph.

### C. Algorithmic pseudocode

#### IV. CONNECTIVITY PRESERVATION GUARANTEE

We establish that the critical communication subgraph stays connected for all time. The guarantee is hybrid: (i) continuous-time link feasibility is enforced by connectivity CBF constraints on the maintained edges, and (ii) discrete-time pruning preserves graph connectivity with a robustness margin against estimation error.

#### A. Continuous-time maintenance of maintained edges via CBF

**Lemma 1.** Fix any undirected edge set  $\mathcal{E}_c \subseteq \mathcal{V} \times \mathcal{V}$  and define

$$\mathcal{C}_{\text{conn}}(\mathcal{E}_c) \triangleq \left\{ x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \leq R_{\max}, \forall (i, j) \in \mathcal{E}_c \right\}.$$

Under the dynamics (2) and Assumption 1–3, suppose that for all  $t \geq 0$  and every  $(i, j) \in \mathcal{E}_c$ , the applied controls satisfy the robust connectivity CBF inequality

$$2z_{ij}^\top(u_i - u_j) \leq \alpha_c(h_{ij}^{\text{conn}}(x)) - 2L_f r_{ij}^2 - 4\bar{d}r_{ij}, \quad (20)$$

where  $z_{ij} = x_i - x_j$ ,  $r_{ij} = \|z_{ij}\|$ , and  $h_{ij}^{\text{conn}}(x) = R_{\max}^2 - r_{ij}^2$ . Then  $\mathcal{C}_{\text{conn}}(\mathcal{E}_c)$  is forward invariant: if  $x(0) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_c)$ , then  $x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_c)$  for all  $t \geq 0$ .

### Algorithm 1 Concurrent Distributed Pruning + CLF-CBF Control (executed by robot $\ell$ )

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1: Init: local estimate  $A^\ell(0)$ , active edge set  $\mathcal{E}(0)$ ,  $k \leftarrow 0$ 
2: while mission not complete do
3:   Communication from neighbors  $\mathcal{N}_\ell(k)$  from (3)–(4)
4:   Exchange  $\{A^p(k)\}_{p \in \mathcal{N}_\ell(k)}$  and local observations
5:   Consensus: update  $A^\ell(k+1)$  via (5)–(6)
6:   Compute  $V_\ell(k)$  by (8); update thresholds  $\varepsilon(k), \mu(k)$  by (9)–(10)
7:   Phase 1 (Proposal): evaluate incident edges  $e = (\ell, j)$ ; build proposal set  $\mathcal{P}_\ell(k)$  using Conditions (i)–(iv)
8:   Phase 2 (Negotiation): exchange proposals; keep only bilaterally approved edges  $\mathcal{A}_\ell(k)$ 
9:   Phase 3 (Conflict): apply deterministic tie-break; output at most one removal  $e_\ell^*(k)$  (or NULL)
10:  Broadcast  $e_\ell^*(k)$ ; update active edge set  $\mathcal{E}(k+1)$  with consistent removals
11:  Set active neighbors  $\mathcal{T}_\ell(k+1) \leftarrow \{j : (\ell, j) \in \mathcal{E}(k+1)\}$ 
12:  for each control step  $t$  until next pruning iteration do
13:    Build safety set  $\mathcal{S}_\ell(t)$ 
14:    Form nominal goal control  $u_\ell^{\text{nom}}(t)$  from CLF (15)
15:    Solve CLF-CBF QP (16)–(19) using  $\mathcal{S}_\ell(t)$  and  $\mathcal{T}_\ell(k+1)$ 
16:    Apply  $u_\ell^*(t)$ ; broadcast  $(x_\ell(t), u_\ell^*(t))$  to neighbors
17:  end for
18:   $k \leftarrow k+1$ 
19: end while

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#### B. Hybrid connectivity preservation under pruning updates

Let  $t_k$  denote the  $k$ -th update time ( $t_{k+1} - t_k = T_d$ ). The maintained topology used by the control layer on  $[t_k, t_{k+1})$  is  $\mathcal{G}_{\text{topo}}(k) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$ . Define the hybrid “connectivity-feasible” set

$$\mathcal{H} \triangleq \left\{ (x, \mathcal{E}) : (\mathcal{V}, \mathcal{E}) \text{ is connected and } x \in \mathcal{C}_{\text{conn}}(\mathcal{E}) \right\}. \quad (21)$$

**Lemma 2.** Fix iteration  $k$  and an edge  $e$  considered for removal. Let  $\mathcal{L}^{\text{true}}(k)$  be the true Laplacian of  $\mathcal{G}_{\text{topo}}(k)$ , and let robot  $\ell$  form an estimate  $\mathcal{L}^\ell(k)$  from its local adjacency estimate  $A^\ell(k)$ . Let  $\Delta \mathcal{L}_e$  denote the Laplacian update associated with removing  $e$ . Define

$$\lambda_{2,\ell}^{\text{pred}}(e, k) \triangleq \lambda_2(\mathcal{L}^\ell(k) - \Delta \mathcal{L}_e), \quad (22)$$

$$\lambda_{2,\ell}^{\text{act}}(e, k) \triangleq \lambda_2(\mathcal{L}^{\text{true}}(k) - \Delta \mathcal{L}_e). \quad (23)$$

Then

$$\lambda_{2,\ell}^{\text{act}}(e, k) \geq \lambda_{2,\ell}^{\text{pred}}(e, k) - \delta_\ell(k). \quad (24)$$

$$\delta_\ell(k) \triangleq \|\mathcal{L}^\ell(k) - \mathcal{L}^{\text{true}}(k)\|_2. \quad (25)$$

*Proof:* Apply Weyl’s inequality to the symmetric matrices  $M_1 = \mathcal{L}^{\text{true}}(k) - \Delta \mathcal{L}_e$  and  $M_2 = \mathcal{L}^\ell(k) - \Delta \mathcal{L}_e$ :

$$|\lambda_2(M_2) - \lambda_2(M_1)| \leq \|M_2 - M_1\|_2 = \|\mathcal{L}^\ell(k) - \mathcal{L}^{\text{true}}(k)\|_2.$$

Rearranging yields (24).

**Theorem 1.** Let  $\lambda_{\min} > 0$  and let  $\mathcal{H}$  be defined in (21). Assume  $(x(t_0), \mathcal{E}_{\text{topo}}(0)) \in \mathcal{H}$  and that the initialization check satisfies  $\lambda_2(\mathcal{L}_0) > \lambda_{\min}$ , where  $\mathcal{L}_0$  is the Laplacian of  $\mathcal{G}_{\text{topo}}(0)$ .

For each update index  $k$ , suppose:

- 1) For all  $t \in [t_k, t_{k+1})$ , the applied controls satisfy (20) for every  $(i, j) \in \mathcal{E}_{\text{topo}}(k)$ .
- 2) At  $t_{k+1}$ , the topology update is the output of the three-phase protocol:

- a) **Phase 1 (Proposal).** Each robot  $\ell$  proposes an incident edge  $e = (\ell, j)$  for removal only if its local prediction satisfies

$$\lambda_{2,\ell}^{\text{pred}}(e, k) = \lambda_2(\mathcal{L}^\ell(k) - \Delta \mathcal{L}_e) \geq \lambda_{\min} + \mu(k), \quad (26)$$

(where  $\mathcal{L}^\ell(k)$  is built from  $\ell$ 's local adjacency estimate  $A^\ell(k)$ ), together with the other local safety checks (redundancy and Lyapunov constraints).

- b) **Phase 2 (Negotiation).** An edge  $e = (i, j)$  is approved only if both endpoints propose it, i.e.,  $e \in \mathcal{P}_i(k) \cap \mathcal{P}_j(k)$ .
- c) **Phase 3 (Conflict resolution).** A deterministic tie-breaking rule selects at most one approved edge  $e^*(k)$  for removal; otherwise no edge is removed. Hence the update satisfies

$$\mathcal{E}_{\text{topo}}(k+1) = \begin{cases} \mathcal{E}_{\text{topo}}(k), & e^*(k) = \text{NULL}, \\ \mathcal{E}_{\text{topo}}(k) \setminus \{e^*(k)\}, & e^*(k) \in \mathcal{E}_{\text{topo}}(k). \end{cases} \quad (27)$$

- 3) Let  $\mathcal{L}^{\text{true}}(k)$  denote the true Laplacian of  $\mathcal{G}_{\text{topo}}(k)$ , and define

$$\delta(k) \triangleq \max_{\ell \in \mathcal{V}} \|\mathcal{L}^\ell(k) - \mathcal{L}^{\text{true}}(k)\|_2.$$

Assume the margin schedule satisfies

$$\mu(k) > \delta(k) \quad \forall k. \quad (28)$$

Then  $\mathcal{H}$  is forward invariant. In particular, for all  $k$  the maintained topology remains connected and maintained links never break:

$$\begin{aligned} &\mathcal{G}_{\text{topo}}(k) \text{ is connected,} \\ &x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_{\text{topo}}(k)), \quad \forall t \in [t_k, t_{k+1}). \end{aligned} \quad (29)$$

*Proof:* Let  $\mathcal{E}_k \triangleq \mathcal{E}_{\text{topo}}(k)$  and let  $\mathcal{L}_k$  be the true Laplacian of  $\mathcal{G}_{\text{topo}}(k) = (\mathcal{V}, \mathcal{E}_k)$  at time  $t_k$ . We prove invariance of  $\mathcal{H}$  across (i) flows and (ii) pruning jumps.

a) *Feasibility on  $[t_k, t_{k+1})$ :* Assume  $(x(t_k), \mathcal{E}_k) \in \mathcal{H}$ . Since  $\mathcal{E}_k$  is constant on  $[t_k, t_{k+1})$  and item (1) holds, Lemma 1 implies

$$x(t_k) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_k) \implies x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_k), \quad \forall t \in [t_k, t_{k+1}), \quad (30)$$

so feasibility is preserved during flows. Graph connectivity of  $(\mathcal{V}, \mathcal{E}_k)$  is unchanged during the interval.

b) *Connectivity at  $t_{k+1}$ :* We show that

$$\lambda_2(\mathcal{L}_k) \geq \lambda_{\min} > 0 \quad \forall k \geq 0, \quad (31)$$

which implies  $\mathcal{G}_{\text{topo}}(k)$  is connected for all  $k$ .

*Base case ( $k = 0$ ).* By the initialization check,  $\lambda_2(\mathcal{L}_0) > \lambda_{\min}$ , as the robots are starting as a connected graph, hence (31) holds at  $k = 0$ .

*Inductive hypothesis.* Assume (31) holds for all indices  $0 \leq j \leq k$ .

*Inductive step (prove for  $k + 1$ ).* By (27), there are two cases:

Case 1: No edge removed. Then  $\mathcal{E}_{k+1} = \mathcal{E}_k$  and  $\mathcal{L}_{k+1} = \mathcal{L}_k$ , so

$$\lambda_2(\mathcal{L}_{k+1}) = \lambda_2(\mathcal{L}_k) \geq \lambda_{\min}.$$

Case 2: One edge  $e^*(k) = (i^*, j^*)$  removed. Then  $\mathcal{E}_{k+1} = \mathcal{E}_k \setminus \{e^*(k)\}$  and

$$\lambda_2(\mathcal{L}_{k+1}) = \lambda_2^{\text{act}}(e^*(k), k).$$

Because  $e^*(k)$  is selected from the approved set, both endpoints proposed it (Phase 2), hence by Phase 1 each endpoint  $\ell \in \{i^*, j^*\}$  verified (26):

$$\lambda_{2,\ell}^{\text{pred}}(e^*(k), k) \geq \lambda_{\min} + \mu(k).$$

Let  $\ell^* \in \{i^*, j^*\}$  attain the minimum predicted value, i.e.,  $\lambda_{2,\ell^*}^{\text{pred}}(e^*(k), k) = \min_{\ell \in \{i^*, j^*\}} \lambda_{2,\ell}^{\text{pred}}(e^*(k), k)$ . Then

$$\lambda_{2,\ell^*}^{\text{pred}}(e^*(k), k) \geq \lambda_{\min} + \mu(k). \quad (32)$$

Applying Lemma 2 with  $\ell = \ell^*$  yields

$$\begin{aligned} \lambda_2(\mathcal{L}_{k+1}) &= \lambda_2^{\text{act}}(e^*(k), k) \\ &\geq \lambda_{2,\ell^*}^{\text{pred}}(e^*(k), k) - \|\mathcal{L}^{\ell^*}(k) - \mathcal{L}^{\text{true}}(k)\|_2 \\ &\geq \lambda_{2,\ell^*}^{\text{pred}}(e^*(k), k) - \delta(k). \end{aligned} \quad (33)$$

Combining with (32) and (28) gives

$$\lambda_2(\mathcal{L}_{k+1}) \geq \lambda_{\min} + \mu(k) - \delta(k) > \lambda_{\min}.$$

Thus (31) holds at  $k + 1$ .

Therefore  $\lambda_2(\mathcal{L}_k) > 0$  for all  $k$ , so  $\mathcal{G}_{\text{topo}}(k)$  is connected for all  $k$ .

c) *Feasibility at  $t_{k+1}$ :* Let  $x^- = x(t_{k+1}^-)$  and note from (30) that  $x^- \in \mathcal{C}_{\text{conn}}(\mathcal{E}_k)$ . At the jump,  $x^+ = x^-$  and  $\mathcal{E}_{k+1} \subseteq \mathcal{E}_k$ , hence

$$x^+ = x^- \in \mathcal{C}_{\text{conn}}(\mathcal{E}_{k+1}),$$

i.e., feasibility is preserved across the pruning jump.

d) *Conclusion:* We have shown feasibility during flows, connectivity at each steps, and feasibility across all the steps; hence  $(x(t), \mathcal{E}_{\text{topo}}(t)) \in \mathcal{H}$  for all  $t \geq t_0$ , i.e.,  $\mathcal{H}$  is forward invariant.

**Remark 1.** The implementation enforces a deterministic tie-breaking so that pruning produces either  $e^*(k) = \text{NULL}$  or a single edge  $e^*(k) \in \mathcal{E}_{\text{topo}}(k)$ , which is exactly the update (27). This assumption is used only to simplify the inductive step.

## V. EXPERIMENTS

TABLE I

SIMULATION SCENARIOS AND STRESS FACTORS USED IN EVALUATION.

| Scenario | $N$ | $R_{\max}$ (m) | $\Delta t$ (s) | Purpose                    |
|----------|-----|----------------|----------------|----------------------------|
| A        | 10  | 3.0            | 0.05           | Baseline                   |
| B        | 10  | 3.5            | 0.10           | Temporal discretization    |
| C        | 10  | 2.0            | 0.03           | Sparse-connectivity stress |
| D        | 20  | 3.0            | 0.05           | Scalability                |
| E        | 12  | 2.2            | 0.08           | Extreme stress test        |

### A. Simulation Setup

We evaluate the proposed *Concurrent Pruning* controller in a 2D underwater environment with spatially-varying flow fields. The experimental setup consists of a  $10 \times 10 \text{ m}^2$  workspace where a fleet of  $n=10$  robots (Scenario A) must navigate from randomly initialized positions to a goal region at  $(8.0, 5.0) \text{ m}$  while maintaining network connectivity under communication radius constraints  $R_{\text{comm}}=3.0 \text{ m}$ .

Robots evolve under discrete-time dynamics with time step  $\Delta t=0.05 \text{ s}$ :

$$x_i[k+1] = x_i[k] + \Delta t \left( f_{\text{flow}}(x_i[k], k\Delta t) + u_i[k] + d_i[k] \right), \quad (34)$$

where  $f_{\text{flow}}(\cdot)$  is the ambient flow field (magnitude  $\approx 0.3 \text{ m/s}$ , spatially varying as shown in Fig. 1),  $u_i[k]$  is the hybrid CLF-CBF control input (bounded by  $\|u_i\| \leq 2.0 \text{ m/s}$ ), and  $d_i[k]$  represents bounded measurement and actuation noise. Safety constraints enforce minimum inter-robot separation  $d_{\text{safe}}=0.3 \text{ m}$ .

The initial topology is the full communication graph, typically yielding  $|E_0| \approx \frac{n(n-1)}{2}$  edges for densely distributed robots. Through concurrent consensus and pruning, the topology converges to a sparse near-tree structure with  $|E_{\text{final}}| \approx n-1$  edges while maintaining algebraic connectivity  $\lambda_2(\mathcal{L}) > 0.2$  throughout. Each simulation runs for 25 seconds, repeated over 10 trials per scenario with different random seeds.

Table I summarizes five test scenarios with varying parameters to validate robustness: Scenario A, B (large time step  $\Delta t=0.2 \text{ s}$ ), C (tight communication radius  $R_{\text{comm}}=2.0 \text{ m}$ ), D (high robot count  $n=20$ ), and E (combined extreme parameters).

*a) Controller execution.*: At every time step, each robot solves a local QP that (i) drives to a common goal  $x_g$  via a CLF constraint and (ii) enforces safety and link-maintenance via CBF constraints (Fig. 1):

- **CLF (goal reaching)**: a Lyapunov function  $V_i(x_i) = \|x_i - x_g\|^2$  is used to generate a stabilizing descent constraint (implemented as a soft constraint in the QP).
- **CBFs (safety + connectivity)**: collision-avoidance CBFs enforce inter-robot separation, and connectivity CBFs enforce  $\|x_i - x_j\| \leq R_{\max}$  on the *maintained* edge set  $\mathcal{E}_{\text{topo}}(k)$  between pruning updates.

Communication updates occur at times  $\{t_k\}$  (Section IV-B), where the maintained graph  $\mathcal{G}_{\text{topo}}(k) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(k))$  may remove at most one edge per update via bilateral approval.

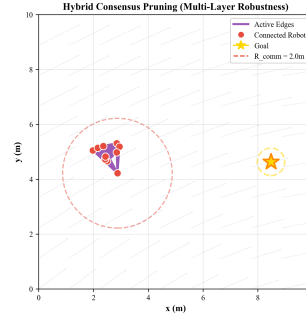


Fig. 1. Illustrative snapshot showing maintained topology: the fleet advances toward the goal while the pruning layer maintains a sparse, connectivity-feasible subgraph.

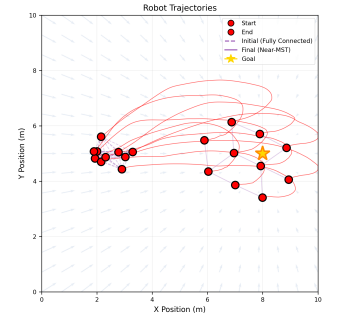


Fig. 2. Robot trajectories (Scenario A) in the flow field with topology evolving from initially dense (dashed) to sparse near-tree (solid).

*b) Baselines.*: We compare four methods, all using the same CLF-CBF motion controller but different maintained topologies: (i) **Concurrent Pruning (ours)**, (ii) **Adjacency Consensus** (sequential pruning), (iii) **Full Graph** where  $\mathcal{E}_{\text{topo}}(k)$  is the instantaneous graph (no edges are explicitly pruned), and (iv) **Centralized MST** (global MST each update).

*c) Success criterion and metrics.*: A trial is deemed successful if: (i) the maintained-topology connectivity satisfies  $\lambda_2(t) \geq \lambda_{\min} = 0.2$  for all  $t$  and (ii) the final goal error is at most  $0.5 \text{ m}$ . We report: (a) success rate, (b) edge reduction relative to the initial topology, (c)  $\min_t \lambda_2(t)$ , (d) cumulative control effort.

## VI. RESULTS

### A. Topology sparsification and connectivity safety

Fig. 3 shows the evolution of the number of maintained edges. The proposed Concurrent Pruning rapidly removes redundant links and stabilizes to a sparse, low-maintenance subgraph across all scenarios. In contrast, the Full-Graph baseline retains substantially more edges, while the centralized MST oracle yields the sparsest topology. The second plot shows the performance of the concurrent pruning method over the five different scenarios.

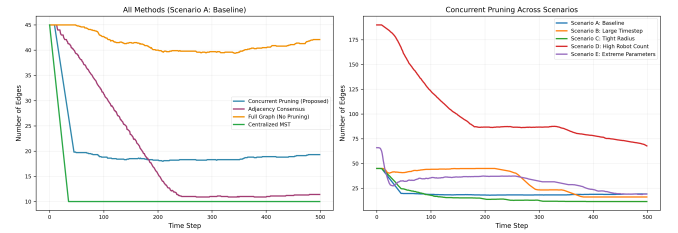


Fig. 3. Edge-count evolution of the maintained topology  $\mathcal{G}_{\text{topo}}(t)$  across different method in same scenario and across scenarios (A–E) with concurrent pruning method.

Fig. 4 reports the corresponding algebraic connectivity  $\lambda_2(t)$  (dashed line: safety threshold). The Full-Graph baseline maintains the largest connectivity margin but with the highest edge count, whereas the MST operates closest to

the threshold, which explains its reduced robustness under extreme stress. Our concurrent pruning method gets to the near MST edges in a smaller timestep than the sequential method which takes way more time to reduce its edge counts while still maintaining comparable algebraic connectivity to it across all scenarios.

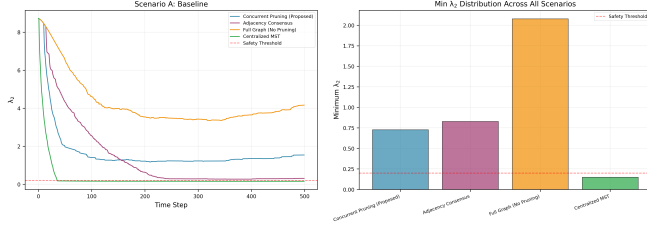


Fig. 4. Algebraic connectivity  $\lambda_2(t)$  of the maintained topology  $\mathcal{G}_{\text{topo}}(t)$  across scenarios (A–E) (dashed:  $\lambda_{\min} = 0.2$ ).

To compactly compare safety across scenarios, Fig. 5 plots the average minimum  $\lambda_2$  per scenario. Concurrent Pruning maintains a conservative margin above  $\lambda_{\min}$  across all scenarios, while MST remains consistently close to the threshold and is therefore more prone to failures when perturbations are severe.

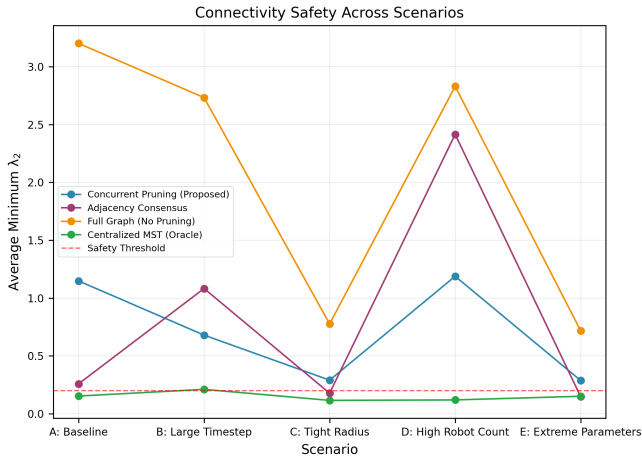


Fig. 5. Connectivity safety across scenarios: average minimum algebraic connectivity (dashed:  $\lambda_{\min} = 0.2$ ).

Fig. 6 reports cumulative control effort. Across scenarios, sparsification does not introduce a prohibitive effort increase relative to dense baselines, indicating that edge reduction primarily lowers communication burden rather than driving large control applications.

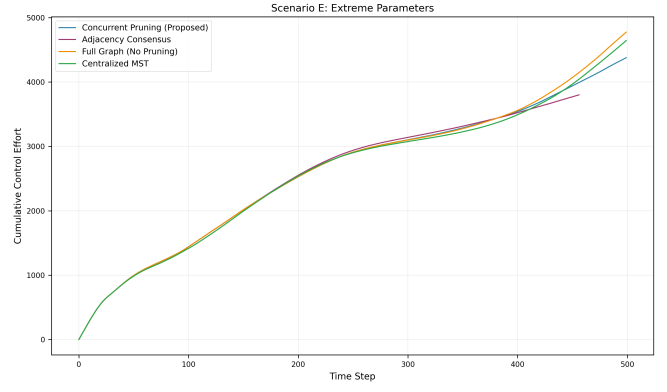


Fig. 6. Cumulative control effort over time. Each panel corresponds to one scenario (A–E).

### B. Success rates, effort, and runtime

Aggregate statistics in Table II quantify the overall trade-off: Concurrent Pruning achieves high reliability (98%) while removing 66% of edges on average, and is substantially faster than the Adjacency-Consensus baseline. The Full-Graph baseline achieves a larger connectivity margin but keeps significantly more edges, and the centralized MST achieves the highest sparsity. (Reported  $\min_t \lambda_2$  averages include both successful and failed trials.)

TABLE II  
AGGREGATE PERFORMANCE ACROSS ALL SCENARIOS (10 MONTE-CARLO TRIALS PER SCENARIO).

| Method              | Success (%) | Edge red. (%)     | $\min_t \lambda_2$  | Runtime (s)       |
|---------------------|-------------|-------------------|---------------------|-------------------|
| Concurrent Pruning  | 97.83       | $66.04 \pm 27.17$ | $0.7276 \pm 1.1044$ | $3.73 \pm 4.95$   |
| Adjacency Consensus | 98.21       | $67.20 \pm 15.47$ | $0.8297 \pm 0.9205$ | $12.91 \pm 17.68$ |
| Full Graph          | 98.78       | $25.48 \pm 20.18$ | $2.0786 \pm 1.1935$ | $1.33 \pm 0.77$   |
| Centralized MST     | 92.02       | $80.57 \pm 4.24$  | $0.1500 \pm 0.0727$ | $1.00 \pm 0.62$   |

## VII. CONCLUSION

This paper presented a fully distributed framework that enables underwater robot fleets to preserve network connectivity while minimizing control effort in flow-dominated environments. By combining distributed adjacency consensus with concurrent edge pruning and integrating it into a CLF–CBF control architecture, the proposed approach achieves sparse, tree-like communication topologies that converge toward minimum spanning structures over time. The concurrent pruning protocol, which uses adaptive thresholds and bilateral negotiation, successfully eliminates redundant communication links while maintaining algebraic connectivity above safety margins while reaching the consensus. Through comprehensive simulations across five stress scenarios with varying communication radii, team sizes, and temporal discretizations, we demonstrated that the method achieves a 98% success rate, reduces edges by 66% on average, and operates 3.5 times faster than sequential pruning alternatives while maintaining comparable connectivity safety. The invariance proof formally guarantees end-to-end connectivity preservation under bounded disturbances and spatially varying flows. These results validate that distributed

teams can rely on minimal information exchange to coordinate efficiently in bandwidth-limited and localization-denied underwater environments. Future work will extend the framework to three-dimensional deployments, incorporate communication delays and packet loss, and validate the approach through field experiments with autonomous underwater vehicles operating in real currents.

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